

Reverse-check review package — GeneralU1_gen

An independent, **blank-slate** agent instance reconstructed the physics below from the sanitized `.fr` alone (no paper, no metadata, no conversation history). A second fresh instance then compared the reconstruction against the source paper. The final verdict belongs to the human reviewer — sign off at the bottom.

item	value
model file	GeneralU1/repair/final.fr
original model name	GeneralU1_gen (hidden from the agent)
paper	GeneralU1/text/2104.10902.txt
blindness scope	blank slate w.r.t. paper, authors, model name, comments and prose labels — NOT w.r.t. field/parameter symbol names, which are kept so the reviewer can map the reconstruction back

Verbatim Lagrangian terms (from the `.fr`)

These are the terms the reconstruction must account for, quoted unmodified. Check each against its reconstructed LaTeX form below.

LZpF (:=)

```
Block[{ff,sp1,sp2,sp3,mu,cc}, -gX Zp[mu] (qQLX
  ↪ uqbar[sp1,ff,cc].Ga[mu,sp1,sp2].ProjM[sp2,sp3].uq[sp3,ff,cc] + qQLX
  ↪ dqbar[sp1,ff,cc].Ga[mu,sp1,sp2].ProjM[sp2,sp3].dq[sp3,ff,cc] + qURX
  ↪ uqbar[sp1,ff,cc].Ga[mu,sp1,sp2].ProjP[sp2,sp3].uq[sp3,ff,cc] + qDRX
  ↪ dqbar[sp1,ff,cc].Ga[mu,sp1,sp2].ProjP[sp2,sp3].dq[sp3,ff,cc] + qLLX
  ↪ vlbar[sp1,ff].Ga[mu,sp1,sp2].ProjM[sp2,sp3].vl[sp3,ff] + qLLX
  ↪ lbar[sp1,ff].Ga[mu,sp1,sp2].ProjM[sp2,sp3].l[sp3,ff] + qERX
  ↪ lbar[sp1,ff].Ga[mu,sp1,sp2].ProjP[sp2,sp3].l[sp3,ff] + qNRX
  ↪ NRbar[sp1,ff].Ga[mu,sp1,sp2].ProjP[sp2,sp3].NR[sp3,ff])]
```

LPhiXKin (:=)

```
Block[{mu}, del[PhiXbar[],mu] del[PhiX[],mu] + I*gX*qPhiX*Zp[mu]*(PhiXbar[] del[PhiX[],mu] -
  ↪ del[PhiXbar[],mu] PhiX[]) + gX^2*qPhiX^2 Zp[mu] Zp[mu] PhiXbar[] PhiX[]]
```

LHiggsX (:=)

```
Block[{ii,mu}, ExpandIndices[I*gX*qHX*Zp[mu]*(Phibar[ii] DC[Phi[ii],mu] - DC[Phibar[ii],mu]
  ↪ Phi[ii]) + gX^2*qHX^2 Zp[mu] Zp[mu] Phibar[ii] Phi[ii], FlavorExpand->{SU2D,SU2W}]]
```

VPhiX (:=)

```
Block[{ii}, ExpandIndices[mPhi2 PhiXbar[] PhiX[] + lamPhi (PhiXbar[] PhiX[])^2 + lamHP
  ↪ (Phibar[ii] Phi[ii]) (PhiXbar[] PhiX[]), FlavorExpand->{SU2D,SU2W}]]
```

LNuDiracYuk (:=)

```
Block[{ff1,ff2,ii,sp1,sp2}, ExpandIndices[-yNu[ff1,ff2]
  ↪ LLbar[sp1,ii,ff1].ProjP[sp1,sp2].NR[sp2,ff2] Phibar[ii], FlavorExpand->SU2D]]
```

LNuMajoranaYuk (:=)

```
Block[{ff,sp1,sp2}, -1/2 yN[ff] PhiX[] NRbar[sp1,ff].ProjP[sp1,sp2].NR[sp2,ff] - 1/2 yN[ff]
  ↪ PhiXbar[] NRbar[sp1,ff].ProjM[sp1,sp2].NR[sp2,ff]]
```

L_{Tot} (:=)

L_{ZpF} + L_{PhiXKin} + L_{HiggsX} - V_{PhiX} + L_{NuDiracYuk} + HC[L_{NuDiracYuk}] + L_{NuMajoranaYuk}

Blank-slate reconstruction

Reconstructed Physics From `sanitized.fr`

Lagrangian

The total Lagrangian defined in the file is

$$\mathcal{L}_{\text{Tot}} = \mathcal{L}_{\text{ZpF}} + \mathcal{L}_{\text{PhiXKin}} + \mathcal{L}_{\text{HiggsX}} - V_{\text{PhiX}} + \mathcal{L}_{\text{NuDiracYuk}} + \mathcal{L}_{\text{NuDiracYuk}}^\dagger + \mathcal{L}_{\text{NuMajoranaYuk}}.$$

The new Abelian covariant derivative convention used by the file is

$$D_\mu = D_\mu^{\text{SM}} + ig_X q_X Z'_\mu,$$

where, for the SM Higgs doublet,

$$D_\mu^{\text{SM}} H = (\partial_\mu + ig W_\mu^a T^a + ig' Y_H B_\mu) H.$$

For the singlet scalar Φ_X , only the new $U(1)_X$ gauge field appears:

$$D_\mu \Phi_X = (\partial_\mu + ig_X q_X^\Phi Z'_\mu) \Phi_X.$$

The chiral projectors are

$$P_L = \frac{1 - \gamma^5}{2}, \quad P_R = \frac{1 + \gamma^5}{2}.$$

L_{ZpF}

$$\begin{aligned} \mathcal{L}_{\text{ZpF}} = & -g_X Z'_\mu \sum_{f=1}^3 \left[q_X^Q (\bar{u}_f \gamma^\mu P_L u_f + \bar{d}_f \gamma^\mu P_L d_f) + q_X^u \bar{u}_f \gamma^\mu P_R u_f + q_X^d \bar{d}_f \gamma^\mu P_R d_f \right. \\ & \left. + q_X^L (\bar{\nu}_f \gamma^\mu P_L \nu_f + \bar{e}_f \gamma^\mu P_L e_f) + q_X^e \bar{e}_f \gamma^\mu P_R e_f + q_X^N \bar{N}_f \gamma^\mu P_R N_f \right]. \end{aligned}$$

Color indices are summed for the quark bilinears. Generation index $f = 1, 2, 3$ is summed.

The $U(1)_X$ charges are defined internally as

$$\begin{aligned} q_X^Q &= \frac{x_H}{6} + \frac{x_\Phi}{3}, & q_X^u &= \frac{2x_H}{3} + \frac{x_\Phi}{3}, & q_X^d &= -\frac{x_H}{3} + \frac{x_\Phi}{3}, \\ q_X^L &= -\frac{x_H}{2} - x_\Phi, & q_X^e &= -x_H - x_\Phi, & q_X^N &= -x_\Phi, \\ q_X^H &= -\frac{x_H}{2}, & q_X^\Phi &= 2x_\Phi. \end{aligned}$$

L_{PhiXKin}

$$\mathcal{L}_{\text{PhiXKin}} = \partial_\mu \Phi_X^\dagger \partial^\mu \Phi_X + ig_X q_X^\Phi Z'_\mu (\Phi_X^\dagger \partial^\mu \Phi_X - \partial^\mu \Phi_X^\dagger \Phi_X) + g_X^2 (q_X^\Phi)^2 Z'_\mu Z'^\mu \Phi_X^\dagger \Phi_X.$$

Equivalently,

$$\mathcal{L}_{\text{PhiXKin}} = (D_\mu \Phi_X)^\dagger D^\mu \Phi_X, \quad D_\mu \Phi_X = (\partial_\mu + ig_X q_X^\Phi Z'_\mu) \Phi_X.$$

The field definition is

$$\Phi_X = \frac{1}{\sqrt{2}} (v_\Phi + \phi_X + iG_{Z'}), \quad v_\Phi = \frac{M_{Z'}}{2g_X}.$$

LHiggsX

$$\mathcal{L}_{\text{HiggsX}} = ig_X q_X^H Z'_\mu [H^\dagger D_{\text{SM}}^\mu H - (D_{\text{SM}}^\mu H)^\dagger H] + g_X^2 (q_X^H)^2 Z'_\mu Z'^\mu H^\dagger H.$$

This is the $U(1)_X$ -dependent part of

$$(D_\mu H)^\dagger D^\mu H, \quad D_\mu H = D_\mu^{\text{SM}} H + ig_X q_X^H Z'_\mu H.$$

VPhiX

The file defines

$$V_{\text{PhiX}} = m_\Phi^2 \Phi_X^\dagger \Phi_X + \lambda_\Phi (\Phi_X^\dagger \Phi_X)^2 + \lambda' (H^\dagger H) (\Phi_X^\dagger \Phi_X).$$

It enters the total Lagrangian as

$$\mathcal{L} \supset -V_{\text{PhiX}}.$$

LNuDiracYuk

$$\mathcal{L}_{\text{NuDiracYuk}} = - \sum_{i,j=1}^3 (Y_\nu)_{ij} \bar{L}_i P_R N_j H^\dagger$$

with the $SU(2)$ index contracted as written in the file:

$$\mathcal{L}_{\text{NuDiracYuk}} = -(Y_\nu)_{ij} \bar{L}_{i,a} P_R N_j H_a^\dagger.$$

The total Lagrangian also includes its Hermitian conjugate:

$$\mathcal{L}_{\text{NuDiracYuk}}^\dagger = -(Y_\nu)_{ij}^* H_a \bar{N}_j P_L L_{i,a}.$$

The file declares Y_ν real, so $Y_\nu^* = Y_\nu$.

LNuMajoranaYuk

$$\mathcal{L}_{\text{NuMajoranaYuk}} = -\frac{1}{2} \sum_{i=1}^3 Y_{N_i} \Phi_X \bar{N}_i P_R N_i - \frac{1}{2} \sum_{i=1}^3 Y_{N_i} \Phi_X^\dagger \bar{N}_i P_L N_i.$$

Equivalently, in two-component chiral notation,

$$\mathcal{L}_{\text{NuMajoranaYuk}} = -\frac{1}{2} \sum_i Y_{N_i} \Phi_X \overline{N_{Ri}^c} N_{Ri} + \text{h.c.}$$

The internal Yukawa definitions are

$$Y_{N_i} = \frac{\sqrt{2} m_{N_i}}{v_\Phi}.$$

Field Table

.fr class	Symbol	Spin	SU(3)	SU(2)	$U(1)$ charge / hyper- charge	Self- conjugate	Mass
V[100]	Zp (Z'_μ)	1	singlet	singlet	gauge boson of $U(1)_X$; no matter charge assigned	yes	$M_{Z'} =$ $M_{Zp} =$ 7500
F[101]	NR (N_1, N_2, N_3)	1/2	singlet	singlet	right- chiral compo- nent has $q_X^N =$ $-x_\Phi$	yes, Majorana	$m_{N_1} =$ $m_{N_2} =$ $m_{N_3} =$ 10000
S[102]	phiX (ϕ_X)	0	singlet	singlet	real exci- tation of Φ_X , whose parent has $X = 2x_\Phi$, $Q = 0$, $Y = 0$	yes	$m_{\phi_X} =$ 1000
S[103]	GZp ($G_{Z'}$)	0	singlet	singlet	Goldstone compo- nent of Φ_X , whose parent has $X = 2x_\Phi$, $Q = 0$, $Y = 0$	yes	$M_{Z'} =$ 7500
S[104]	PhiX (Φ_X)	0	singlet	singlet	$Q =$ 0 , $Y =$ 0 , $X =$ $2x_\Phi$	no	unphysical field, $\Phi_X =$ ($v_\Phi +$ $\phi_X +$ $iG_{Z'})/\sqrt{2}$

Parameters

Parameter	Value in file	Multiplies / controls	Physical meaning
gX	0.1	every Z' interaction and $U(1)_X$ covariant derivative	new Abelian gauge coupling
xH	0	internal charge definitions $q_X^Q, q_X^u, q_X^d, q_X^L, q_X^e, q_X^H$	coefficient controlling the hypercharge-like part of the $U(1)_X$ charge assignment

Parameter	Value in file	Multiplies / controls	Physical meaning
xPhi	1	internal charge definitions, especially $q_X^\Phi = 2x_\Phi$ and $q_X^N = -x_\Phi$	coefficient controlling the singlet-scalar and right-handed-neutrino $U(1)_X$ charge normalization
lamPhi	0.1	$(\Phi_X^\dagger \Phi_X)^2$ in VPhiX	quartic self-coupling of the new complex singlet scalar
lamHP	0	$(H^\dagger H)(\Phi_X^\dagger \Phi_X)$ in VPhiX	Higgs-portal quartic coupling
mPhi2	1000000	$\Phi_X^\dagger \Phi_X$ in VPhiX	quadratic scalar-potential mass parameter for Φ_X
yNu[i,j]	all entries 0	$-(Y_\nu)_{ij} \bar{L}_i P_R N_j H^\dagger + \text{h.c.}$	Dirac neutrino Yukawa matrix coupling SM lepton doublets to the new Majorana singlets

The file also defines the following internal parameters:

$$v_\Phi = \frac{M_{Z'}}{2g_X}, \quad Y_{N_i} = \frac{\sqrt{2} m_{N_i}}{v_\Phi}.$$

Physics Summary

The file encodes a $U(1)_X$ extension of the Standard Model with a new neutral gauge boson Z' , three self-conjugate singlet fermions N_i , and a complex singlet scalar Φ_X whose vacuum expectation value breaks the new gauge symmetry and supplies the Z' and N_i masses. The Z' couples chirally to SM quarks, charged leptons, neutrinos, and the right-handed component of the Majorana singlets with charges determined by x_H and x_Φ .

It mediates dilepton, dijet, neutrino, and heavy-neutral-lepton production through s -channel Z' exchange, while the scalar sector allows singlet-Higgs portal interactions and Majorana mass generation through $\Phi_X \bar{N}_R^c N_R$.

Paper cross-check

Comparison of reconstruction.md Against the Paper Model

Paper Model Location

The paper defines the minimal $U(1)_X$ model in **Sec. II, “The $U(1)_X$ Model”**. The relevant definitions are:

- **Table I:** gauge representations and $U(1)_X$ charges of $q_L, u_R, d_R, \ell_L, e_R, N_R, H, \Phi$.
- **Eq. (1):** Yukawa interactions.
- **Eq. (2):** renormalizable Higgs potential.
- **Eq. (3):** scalar VEVs.
- **Eq. (4):** Z' mass after symmetry breaking.
- **Eq. (6):** Z' interactions with SM fermions.
- **Eqs. (7)-(9):** partial widths, including light neutrino and RHN couplings.

Term-by-Term Comparison

paper term (eq. ref)	reconstruction term	verdict	notes
Gauge group $SU(3)_C \otimes SU(2)_L \otimes U(1)_Y \otimes U(1)_X$, Sec. II	Same model summarized as a $U(1)_X$ extension	agree	Same gauge extension.
Field content and reps for SM fermions, N_R , H , Φ , Table I	New-field table plus $U(1)_X$ charge formulas for SM fermions	missing-in-reconstruction	Reconstruction does not give the full SM field representation table under $SU(3)_C$, $SU(2)_L$, $U(1)_Y$, though it does encode their $U(1)_X$ charges in the Z' couplings.
$q_X(q_L) = \frac{x_H}{6} + \frac{x_\Phi}{3}$, Table I	$q_X^Q = \frac{x_H}{6} + \frac{x_\Phi}{3}$	agree	Matches.
$q_X(u_R) = \frac{2x_H}{3} + \frac{x_\Phi}{3}$, Table I	$q_X^u = \frac{2x_H}{3} + \frac{x_\Phi}{3}$	agree	Matches.
$q_X(d_R) = -\frac{x_H}{3} + \frac{x_\Phi}{3}$, Table I	$q_X^d = -\frac{x_H}{3} + \frac{x_\Phi}{3}$	agree	Matches.
$q_X(\ell_L) = -\frac{x_H}{2} - x_\Phi$, Table I	$q_X^L = -\frac{x_H}{2} - x_\Phi$	agree	Matches.
$q_X(e_R) = -x_H - x_\Phi$, Table I	$q_X^e = -x_H - x_\Phi$	agree	Matches.
$q_X(N_R) = -x_\Phi$, Table I	$q_X^N = -x_\Phi$	agree	Matches.
$q_X(H) = -\frac{x_H}{2}$, Table I	$q_X^H = -\frac{x_H}{2}$	agree	Matches.
$q_X(\Phi) = 2x_\Phi$, Table I	$q_X^\Phi = 2x_\Phi$	agree	Matches.
Full Yukawa sector, Eq. (1): quark and charged-lepton Yukawas plus neutrino Yukawas	Reconstruction includes only neutrino Dirac and Majorana Yukawas	missing-in-reconstruction	The paper's SM Yukawa terms Y_u, Y_d, Y_e are absent. This may be intentional if the implementation only reconstructs the new-sector additions, but it is not the full paper Lagrangian.
Dirac neutrino Yukawa, Eq. (1): $-Y_\nu^{\alpha\beta} \ell_L^\alpha H N_R^\beta + \text{H.c.}$, as printed	$-(Y_\nu)_{ij} \bar{L}_{i,a} P_R N_j H_a^\dagger + \text{H.c.}$	disagree	The reconstruction uses $H^\dagger$. With Table I charges, the gauge-invariant $U(1)_X$ contraction uses the Higgs field carrying $q_X(H) = -x_H/2$, not its conjugate. The reconstruction term is only $U(1)_X$ -neutral for special choices such as $x_H = 0$.
Majorana Yukawa, Eq. (1): $-Y_N^\alpha \Phi N_R^{\alpha c} N_R^\alpha + \text{H.c.}$	$-\frac{1}{2} Y_{N_i} \Phi_X \bar{N} P_R N - \frac{1}{2} Y_{N_i} \Phi_X^\dagger \bar{N} P_L N$	agree	Same physics in four-component Majorana notation. The factor 1/2 is conventional for Majorana four-component writing.

paper term (eq. ref)	reconstruction term	verdict	notes
Higgs potential, Eq. (2): $-m_h^2 H^\dagger H + \lambda(H^\dagger H)^2 + m_\Phi^2 \Phi^\dagger \Phi + \lambda_\Phi(\Phi^\dagger \Phi)^2 + \lambda'(H^\dagger H)(\Phi^\dagger \Phi)$	$m_\Phi^2 \Phi_X^\dagger \Phi_X + \lambda_\Phi(\Phi_X^\dagger \Phi_X)^2 + \lambda'(H^\dagger H)(\Phi_X^\dagger \Phi_X),$ entering as $-V$	missing-in-reconstruction	The Φ mass, Φ quartic, and portal terms agree, but the SM Higgs potential terms $-m_h^2 H^\dagger H + \lambda(H^\dagger H)^2$ are missing.
Scalar VEVs, Eq. (3): $\langle H \rangle$ and $\langle \Phi \rangle = (v_\Phi + \phi)/\sqrt{2}$	$\Phi_X = (v_\Phi + \phi_X + iG_{Z'})/\sqrt{2}$	agree	The Φ expansion agrees up to inclusion of the Goldstone mode. The reconstruction does not state the Higgs VEV.
Higgs VEV, Eq. (3): $\langle H \rangle$ with electroweak VEV $v \simeq 246$ GeV	Not included	missing-in-reconstruction	The reconstruction uses H in interactions but omits the Higgs VEV definition.
Z' mass, Eq. (4): $M_{Z'} = \frac{g'}{2g'v_\Phi} \sqrt{4v_\Phi^2 + \frac{1}{4}x_H^2 v^2} \simeq 2g'v_\Phi$	$v_\Phi = M_{Z'}/(2g_X)$	agree	Agrees with the paper's large- v_Φ , $x_\Phi = 1$ approximation. The reconstruction does not retain the small electroweak contribution proportional to $x_H^2 v^2$.
Z' couplings to SM quarks and leptons, Eq. (6)	$-g_X Z'_\mu \sum_f [\dots]$ with P_L, P_R charges	agree	Same chiral charge structure, modulo notation $g_X \leftrightarrow g'$.
Z' coupling to light neutrinos, implied by Table I and Eq. (8)	$q_X^L \bar{\nu} \gamma^\mu P_L \nu$	agree	Same left-handed light-neutrino coupling.
Z' coupling to RHNs, Table I and Eq. (9)	$q_X^N \bar{N} \gamma^\mu P_R N$	agree	Same right-chiral RHN charge. In Majorana notation this corresponds to the usual chiral/axial current.
Φ kinetic term implied by gauge symmetry and used in Eq. (4)	$(D_\mu \Phi_X)^\dagger D^\mu \Phi_X$ expanded	agree	Correct covariant derivative structure with $q_X^\Phi = 2x_\Phi$.
Higgs $U(1)_X$ covariant kinetic contribution implied by Table I and Eq. (4)	$U(1)_X$ -dependent part of $(D_\mu H)^\dagger D^\mu H$	agree	Correct $q_X^H = -x_H/2$ dependence.
Explicit Goldstone field $G_{Z'}$ and unphysical Φ_X component fields	Included in field table	extra-in-reconstruction	The paper writes only the physical singlet fluctuation ϕ in Eq. (3). The Goldstone is an implementation-level gauge-basis detail, not a physics conflict.

paper term (eq. ref)	reconstruction term	verdict	notes
Fixed parameter values $g_X = 0.1$, $x_H = 0$, $x_\Phi = 1$, $M_{Z'} = 7500$, $m_N = 10000$, $m_{\phi_X} = 1000$	Listed as file values	extra-in-reconstruction	The paper fixes $x_\Phi = 1$ without loss of generality and often uses $M_{Z'} = 7.5$ TeV benchmarks, but it scans x_H and commonly uses different g' values. These are implementation benchmark choices, not general paper definitions.

Disagreements and Severity

issue	severity	what a human should check
Dirac neutrino Yukawa uses $H^\dagger$ in the reconstruction instead of the Higgs contraction appearing in Eq. (1).	substantive	Check the original implementation's $SU(2)$, hypercharge, and $U(1)_X$ conventions, because with Table I charges the reconstructed $H^\dagger$ term is not gauge invariant for general x_H .
SM quark and charged-lepton Yukawa terms from Eq. (1) are absent.	substantive	Check whether the implementation intentionally reconstructed only BSM additions or was meant to encode the full minimal $U(1)_X$ Lagrangian.
SM Higgs potential terms $-m_h^2 H^\dagger H + \lambda(H^\dagger H)^2$ from Eq. (2) are absent.	substantive	Check whether the base SM Higgs sector is imported elsewhere in the implementation or accidentally omitted.
Full SM field representation table from Table I is not reproduced.	cosmetic	Check whether the reconstruction was intended to document only new fields and derived $U(1)_X$ charges rather than the full model content.
Higgs VEV from Eq. (3) is omitted.	convention	Check whether electroweak symmetry breaking is handled by an external SM model file.
Z' mass relation uses $v_\Phi = M_{Z'}/(2g_X)$, dropping the $x_H^2 v^2/4$ contribution in Eq. (4).	convention	Check whether the implementation is explicitly working in the paper's $v_\Phi^2 \gg v^2$, $x_\Phi = 1$ approximation.
Reconstruction includes explicit $G_{Z'}$ and unphysical Φ_X fields not shown in the paper's Eq. (3).	cosmetic	Check the gauge choice in the implementation; this is likely just an implementation-basis detail.

issue	severity	what a human should check
Reconstruction lists fixed benchmark parameters such as $g_X = 0.1$, $x_H = 0$, and $m_{\phi_X} = 1000$.	convention	Check whether these are harmless numerical defaults or whether they restrict scans that the paper treats as free parameters.

Overall Assessment

The reconstruction captures the central $U(1)_X$ charge assignment, the chiral Z' couplings, the singlet-scalar kinetic structure, the Φ -sector potential terms, and the RHN Majorana Yukawa structure of the paper’s Sec. II model. The most important physics mismatch is the reconstructed Dirac neutrino Yukawa contraction with $H^\dagger$, which conflicts with the paper’s Table I charge assignments for general x_H . Several other omissions appear to be boundary choices between a BSM implementation file and a full model definition: the SM Yukawas, SM Higgs potential, SM field reps, and Higgs VEV may live in an imported SM sector, but they are part of the paper’s complete Lagrangian as written.

Suggested checks for the reviewer

1. Every verbatim `.fr` term above has a reconstructed LaTeX counterpart with the same field content, chirality and conjugation.
2. Kinetic terms: covariant derivative gauge content matches the field’s representations; normalization is canonical.
3. Non-self-conjugate interaction terms appear together with their Hermitian conjugates.
4. Quantum numbers in the field table match the `.fr` declarations (and the paper, where the cross-check table flags disagreements).
5. Numeric masses/couplings are placeholders unless the paper pins them — treat values as demo inputs, not measurements.
6. Sanitizer scope: 20 prose labels were scrubbed; field/parameter symbol names were kept and may hint at the model’s identity.

Physicist sign-off

- Reviewed by: _____ Date: _____
- Verdict: [] approve [] approve with corrections [] reject
- Notes: